ACU Routing API — IT Deployment & Maintenance Guide

This document contains everything the IT department needs to deploy, maintain, and update the ACU Routing API on a bare-metal Windows Server with minimal friction.

1. Initial Setup (One-Time)

To set up the API on a new Windows Server:

1. **Prerequisites**: Ensure the server has **Python 3.12+** and **PostgreSQL** installed.
2. **Transfer Files**: Copy the entire `centralized_routing_api` folder to the server.
3. **Run the Wizard**: Open **PowerShell as Administrator**, navigate to the project folder, and run:

   ```
   .\scripts\setup_server.ps1
   ```

4. **Follow Prompts**: The script will automatically ask for the database password and create an initial Administrator account. It will then automatically:
   - Create the `.env` configuration file
   - Install all Python dependencies
   - Safely create the database and required tables
   - Register the API as an auto-restarting Windows Background Service (`ACURoutingAPI`)
   - Register automated Database Backup Tasks in Windows Task Scheduler

2. General Maintenance

Once the setup wizard finishes, the API runs entirely in the background via Waitress (a multi-threaded production server).

- **Service Name**: `ACURoutingAPI`
- **Port**: `8080` (Ensure this port is open in the Windows Defender Firewall)
- **Auto-Recovery**: If the API crashes for any reason, Windows will automatically restart it within 5 seconds.
- **Logs**: All access and error logs are automatically written to the `logs\` folder inside the project directory. They rotate daily to prevent disk fill-up.

Stopping/Starting the Service

Open PowerShell as Administrator:

- To stop: `Stop-Service ACURoutingAPI`
- To start: `Start-Service ACURoutingAPI`
- To restart: `Restart-Service ACURoutingAPI`

3. Database Backups (Fully Automated)

The setup wizard (`setup_server.ps1`) automatically configured two backup tasks in Windows Task Scheduler to secure your data without any manual intervention.

- **Weekly Backup**: Runs every Sunday at 2:00 AM.
- **Monthly Backup**: Runs on the 1st of every month at 3:00 AM.
- *Location*: All backups are compressed into a custom PostgreSQL format (`.dump`) and saved securely inside the `backup\` folder in the project directory.

Smart Cleanup (Retention Policy): The backup scripts are self-cleaning to prevent disk fill-up:

- Weekly backups are automatically deleted after 90 days.
- Monthly backups are automatically deleted after 360 days.

4. Best Practices for Applying Updates

When the development team provides an update (new code), follow these strict steps to ensure zero data loss and a clean rollout:

1. **Stop the Service** *Always stop the API before replacing files so files aren't locked.*

   ```
   Stop-Service ACURoutingAPI
   ```

2. **Replace the Code** Copy the new files over the existing ones. **CRITICAL**: *Do not overwrite the `.env` file or the `logs\` directory.* The `.env` file contains the server's unique database credentials and secure tokens.

3. **Install Any New Dependencies** If the update includes new libraries, run the dependency installer:

   ```
   python -m pip install -r requirements.txt
   ```

4. **Restart the Service**

   ```
   Start-Service ACURoutingAPI
   ```

5. **Verify** Check `logs\api_stdout.log` to ensure the server started successfully without throwing errors. You can also visit `http://localhost:8080/api/health` to verify liveness.